

Ethical AI & Challenges

Student study guide

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Your take-home companion to the session. The slides carry the key ideas; this guide carries the explanation behind each one. Read it after class to consolidate, or before an assessment to revise.

What you should be able to do after this session

- Recognise where AI can quietly harm people — the four doors.
- Explain why the responsibility for an AI-influenced decision stays with you.
- Ask the five questions of any AI decision.
- Name concrete habits you can start on Monday.

The one idea to carry out of the room: *AI amplifies a decision — it never owns it. The duty to check for harm, to explain, and to protect people stays with the human closest to the decision: you.*

1. The hook: "a computer can never be held accountable"

An IBM internal training manual from **1979** stated: "*A computer can never be held accountable. Therefore a computer must never make a management decision.*" It was decades ahead of its time. A computer cannot be blamed, cannot apologise, cannot answer to anyone — so responsibility for what it does can never rest with the machine. It stays with the human.

That is the whole session in one line: when AI helps make a decision, a person still owns it.

Source: IBM internal training manual, 1979.

2. Why this is yours

AI does not create judgment — it **amplifies** it, repeating the patterns in its training data faster and at larger scale. So when an AI-shaped decision harms someone, the responsibility flows to the person closest to the decision: whoever used the tool, approved its output, or failed to question it.

"The algorithm said so" is not a defence — it is an *abdication*, using the tool's opacity as a substitute for your own judgment. Courts, regulators, and the affected person all reject it.

The trust gap. Organisations have adopted AI widely, but few people trust it to make *fair, unbiased* decisions. That gap is not a public-relations problem — it is precisely where your judgment has to do its work, as the check between the tool and the person.

Why it matters: this session is not about becoming a data scientist or a lawyer. It is about the ordinary moments when an AI answer lands on your desk and you decide what to do next. Those are ethical decisions, whether or not they feel like it.

3. The four doors: where ethical harm enters

Almost every ethical problem with AI enters through one of four doors. Learn the map, then ask of any situation: *which door is open here?*

Door	The question it raises
Bias	Is the AI repeating past discrimination — at scale?
Opacity	Can you explain the decision to the person it affects?
Privacy & consent	Do you know what the tool does with people's data?
Misuse & hallucination	Is the AI trusted beyond what it can actually do?

4. Door 1 — Bias: trained on our past, repeated at scale

Bias needs no bad intent — only unchecked assumptions in the data.

- **Amazon (scrapped 2018).** A hiring AI trained on ten years of mostly-male hiring learned to penalise CVs containing "women's" (e.g. "women's chess club") and downgraded all-women colleges.
- **Mobley v. Workday (live, 2025–26).** A class action alleges Workday's AI screening discriminated by race, age, and disability. In **March 2026 a judge allowed the age-discrimination claims to proceed** — and signalled that anti-discrimination law can reach the AI *vendor* and protects job *applicants*, not only current employees.
- **ACLU v. Intuit / HireVue (2025).** A deaf, Indigenous applicant was rejected after an AI video interview and told to "practice active listening" — a bias *and* accessibility failure landing on a real person.
- **iTutorGroup (EEOC settlement, 2023).** AI recruiting software auto-rejected women 55+ and men 60+; 200+ qualified applicants rejected — the first EEOC AI-discrimination settlement.

Your leverage: most of what *produces* bias is hidden inside the model, but most of what *reveals* it — the pattern in outcomes over time — is visible to you. Watch the outcomes.

5. Door 1b — Bias is often a measurement problem (the proxy trap)

A US healthcare algorithm (Obermeyer et al., *Science*, 2019) used **healthcare cost** as a stand-in for **health need**. Because less had historically been spent on Black patients at the same level of illness, the model judged them healthier than they were and under-flagged them. Only **17.7%** of patients flagged for extra care were Black; correcting the proxy raised it to **46.5%**.

The tool looked technically sound — it just measured the wrong thing (*spending*, not *sickness*). Every AI tool uses a proxy. Always ask: **what is this actually measuring, and is that the thing we care about?**

6. Door 2 — Opacity: "the algorithm decided" is not an explanation

When someone asks "*why was I rated low?*", you owe them a real answer. But most AI tools people use **cannot explain their own reasoning**:

- *Interpretability* = the logic is understandable by design (most tools aren't).
- *Explainability* = reasons added afterward — approximations, not the model's actual reasoning.
- **The Clever Hans effect**: an AI can get the right answer for the wrong reason. A correct result does not prove correct logic.

A decision you cannot explain is a decision you should not have made on AI output alone.

7. Door 3 — Privacy & consent: know what the tool does with data

Before feeding a tool personal data, know: **what it collects, who can access it, how long it is kept, and whether consent was meaningful**. Centralised procurement does not absolve the person at the keyboard.

Shadow AI is the everyday version: people paste confidential or personal data into public tools because the approved options are inadequate and the pressure is real. Data pasted into a public tool may be stored, used for training, and **cannot be taken back**. (Samsung engineers leaked proprietary source code into ChatGPT this way in 2023.)

8. Door 4 — Misuse & hallucination: trusting AI beyond its limits

Generative AI produces fluent, confident content that can be **flatly wrong** — invented facts, fake citations, misstated figures — and it looks exactly like reliable output. A 2024 enterprise survey found **47%** of business AI users made a major decision based on hallucinated content.

The discipline: treat AI output as a **first draft that needs human verification**, never a finished product that only needs formatting — especially before it reaches someone it affects.

9. The human factor: automation bias

The harms above are structural; what makes them land on real people is usually *how we use the tool*.

- **Automation bias** — over-trusting the AI's answer even when your own knowledge says it's wrong.
 - **The order effect** — if you read the AI's answer *first*, you anchor to it, and everything after is rationalisation. **Forming your own view first** is the cheapest protection you have.
 - **Two failure modes** — *commission* (acting on bad advice you should have rejected) and *omission* (missing a problem the AI didn't flag).
 - **Who's most exposed** — early-career people: familiar enough to trust the tool, not experienced enough to catch it. Time pressure makes it worse for everyone.
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10. The five questions for any AI decision

The practical takeaway. These work in any of the four doors:

1. **Who does this affect — and did they know / consent?**
 2. **What is this tool actually measuring — and is that what we care about?**
 3. **Can I explain this decision to the person it affects?**
 4. **Did I form my own view before I looked at the AI's answer?**
 5. **If this goes wrong, what is the path to fix it?**
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11. What you do differently on Monday

No new tools, no budget, no permission:

1. **Form your own view first** — a two-minute independent take before you read the AI's answer breaks the anchoring effect.
 2. **Keep a plain decision note** — what the AI recommended, what you decided, and why if they differ. Over months it becomes your early-warning record for bias and drift.
 3. **Never deliver an outcome as "the system said so"** — build the human explanation first. Legitimate: *"The AI tool contributed to this — let me walk you through the factors."* Abdication: *"The system scored it that way, I don't have more detail."*
 4. **Verify before it reaches a person** — AI output is always a first draft.
 5. **Tell people their right to a human review** — under GDPR and the EU AI Act, they can contest solely-automated decisions with significant effects. Most don't know it.
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12. The law is catching up

- **GDPR (Art. 22)** — a right not to be subject to solely-automated decisions with significant effects, and (per the EU Court of Justice, 2024) a right to a genuinely intelligible explanation.
 - **EU AI Act** — hiring, performance, and task-allocation AI are **high-risk**, requiring human oversight and meaningful explanations. High-risk penalties up to **€15M / 3%** of global turnover (the €35M/7% tier is for *prohibited* practices). Most stand-alone high-risk obligations were deferred to **December 2027** (June 2026 Digital Omnibus).
 - **US (EEOC)** — existing anti-discrimination law (Title VII, ADEA) applies fully to AI employment decisions; "the algorithm did it" is no defence. Live cases: *Mobley v. Workday*, *Harper v. Sirius XM*.
 - **Colorado AI Act** — substantially **amended (SB 26-189, May 2026)** and delayed to **Jan 2027**: a narrower regime requiring notice, a human-review/adverse-action process, and record-keeping, enforced only by the state Attorney General.
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13. The line to remember

Ethical AI isn't a policy to read — it's a call you make. Every time an AI output reaches your desk, the ethics of what happens next are yours.

Glossary

- **Amplification** — AI repeats and scales the patterns in its training data; it does not create new judgment.
 - **Bias** — systematically unfair outcomes for a group, usually from assumptions baked into training data.
 - **Proxy** — a measurable stand-in for the thing you actually care about (e.g. *cost* as a proxy for *health need*). Proxies are where hidden bias often lives.
 - **Interpretability vs explainability** — logic understandable by design vs. reasons added after the fact.
 - **Automation bias** — over-trusting automated output and under-weighting your own judgment.
 - **Hallucination** — confident, fluent, but false AI output.
 - **Shadow AI** — use of unapproved AI tools, often with sensitive data, outside organisational oversight.
 - **GDPR Article 22** — the EU right regarding solely-automated decisions with significant effects.
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Sources & to go deeper

- **Cases** — Amazon (2018); *Mobley v. Workday* (2025–26); *Harper v. Sirius XM* (2025); ACLU v. Intuit/HireVue (2025); EEOC v. iTutorGroup (2023).
- **Research** — Obermeyer et al. (2019), *Science*; automation-bias literature (Mosier & Skitka); 2024 enterprise hallucination survey.
- **Law** — GDPR Art. 22 (+ EU Court of Justice 2024); EU AI Act; EEOC guidance; Colorado AI Act (SB 26-189, 2026).

This guide is the student companion to `ethical-ai-challenges-deck.md`. For the full instructor material see `reference.md`.